

System & UI Design

Draft for architect/product approval

Safer scheduling. Clearer care.

A responsive appointment experience designed around zero confirmed double-bookings, fast availability search and protected patient data.

- 11 design frames
- 4 user roles
- Desktop + mobile

STATUS
Draft for architect and product designer approval

Document map

01	Cover & Traceability	07	Manager — Desktop
02	System Design	08	Patient — Mobile
03	Foundations & Components	09	Reception — Mobile
04	Patient — Desktop	10	Doctor — Mobile
05	Reception — Desktop	11	Manager — Mobile
06	Doctor — Desktop		

Traceability anchors

DRV-001	Consistency / ADR-002 Transactional unique slot constraint
DRV-002	Performance / ADR-003 Read-optimized availability projection
DRV-003	Security / ADR-004 Role and ownership checks + audit trail
ADR-005	Reliable notifications Outbox and provider adapter

System Design

C4 views, modular boundaries, domain model, booking sequence and operational safeguards.

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Context & trust boundaries

C4 · Context



API authorization is the trust boundary
Role, ownership and audit checks protect confidential patient and appointment data.

Containers & data flow

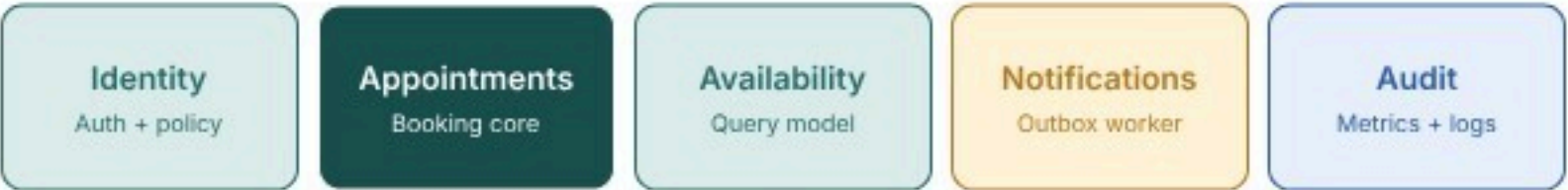
C4 · Container



ADR-001 modular monolith · ADR-002 transaction · ADR-003 projection · ADR-005 outbox

Module & domain boundaries

Components + domain



Clinic 1 → * Doctor Doctor 1 → * Slot Patient 1 → * Appointment

Appointment → NotificationRequest · No EMR data in MVP · Payment and telemedicine remain extension points.

Book appointment sequence

QA-001 · ADR-002

- 1 User selects available slot
- 2 Web → POST /appointments
- 3 API inserts appointment with unique slot key
- 4a Commit → AppointmentBooked + NotificationRequested
- 4b Constraint violation → 409 slot unavailable
- 5 Web confirms or asks user to refresh

GET /availability · POST /appointments · PATCH /appointments/{id} · GET /doctor-schedule

Foundations & Components

Reusable primitives aligned to the approved implementation design system.

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Color tokens

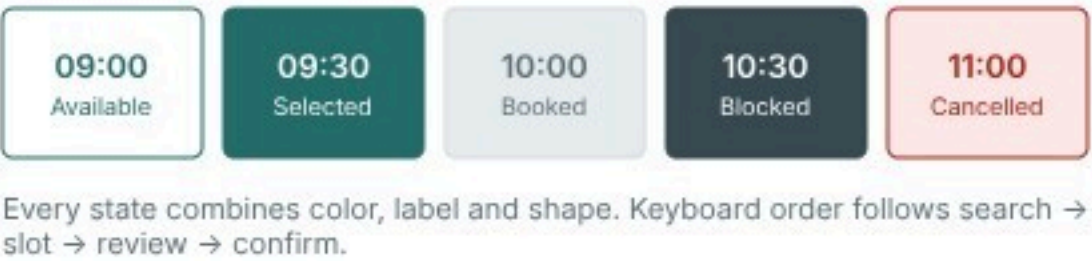


Type & spacing

Page title · 24/600
Section title · 18/600
Body text · 14/400
Caption · 12/400

4px 8px 12px 16px 24px

Accessible state language



Buttons



Inputs & filters



Schedule row



Feedback



Operational metric



Availability slot



Find an appointment

Search, select and confirm in one short flow.

Keyboard-accessible slots

Clinic

Specialty

Doctor

Date

Search

12 available times

Updated just now

09:00	Dr. Maya Patel Dermatology · Downtown	Available
10:30	Dr. Maya Patel Dermatology · Downtown	Selected
13:00	Dr. Noah Kim Dermatology · Northside	Available

No results is distinct from a failed search; change filters or try another date.

Review booking

Dr. Maya Patel

Dermatology · Downtown Clinic
Tuesday, July 14 · 10:30–11:00

09:00 Available	10:30 Selected	11:00 Booked
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Confirm appointment

Reminder pending
Booking is confirmed while email/SMS retries.

Upcoming appointment

Jul 14 · 10:30 · Dr. Maya Patel

Reschedule Cancel

Slot no longer available
Refresh availability and choose another time.

Daily schedule

Scan appointments, manage doctor calendars and resolve conflicts.

+ New appointment

ClinicDoctorDate

Downtown

Dr. Maya Patel

Jul 14, 2026

18 appointments2 open slots1 notification pending

Tuesday, July 14

Compact operational view

08:30	Jordan Lee · New patient Room 2 · jordan@example.com	Confirmed
09:00	Sofia Rivera · Follow-up Room 2 · SMS reminder pending	Pending
09:30	BLOCKED · Team meeting Doctor unavailable	Blocked
10:00	Amir Hassan · Follow-up Room 2 · 555-0104	Confirmed
10:30	Open appointment slot 30 minutes · Bookable	Available

Manage appointment

Patient

Sofia Rivera

DoctorDate

Dr. Maya Patel

Jul 14

10:30
Selected

11:00
Available

11:30
Booked

Save changes

Reschedule

×

Schedule conflict
10:30 was just booked. Refresh and choose another time.

Block doctor availability

FromTo

15:00

16:00

Reason

Clinic meeting

Add block

My schedule

An accurate daily view with operational details only.

< Previous

Tuesday, July 14

Next >

8 appointments

Downtown Clinic

08:30

Jordan Lee · New patient

Room 2 · 30 min

Checked in

09:00

Sofia Rivera · Follow-up

Room 2 · 30 min

Confirmed

09:30

Availability block

Team meeting

Blocked

10:00

Amir Hassan · Follow-up

Room 2 · 30 min

Confirmed

11:00

Taylor Chen · Follow-up

Room 3 · 30 min

Confirmed

Privacy by design

Only operational appointment details are shown; no medical records.

Checked in

Jordan Lee

New patient · Tuesday, July 14 · 08:30–09:00
Room 2 · jordan@example.com

Operational note

Patient requested accessible entrance directions. No clinical notes are stored here.

✓ Mark completed

Report issue

Schedule unavailable

Retry without exposing stale patient details.

Appointment overview

Consolidated operational visibility without medical-record exposure.

Clinic

Date range

All clinics

Jul 1–31, 2026

Apply filters

Appointments

1,284

+8% this month

Completed

1,071

83% completion

No-shows

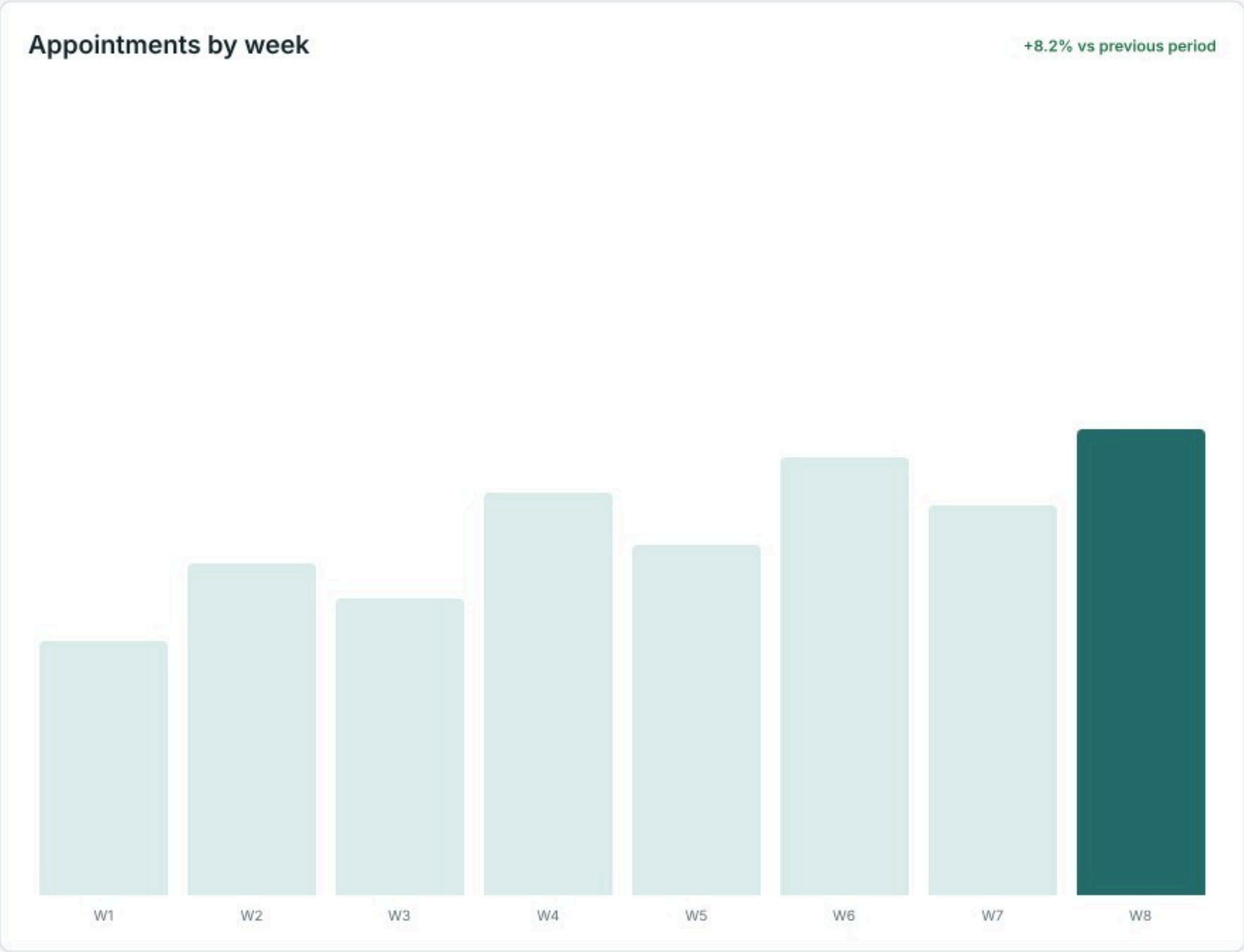
76

5.9% of visits

Pending retries

14

3 provider failures



No-show monitoring

6.8% Downtown Clinic

42 no-shows · 615 appointments

Needs review

5.2% Northside Clinic

21 no-shows · 402 appointments

On track

4.9% West Clinic

13 no-shows · 267 appointments

On track

Notification reliability

98.7% delivered

14 pending retries

3 provider failures

Provider outages never roll back confirmed bookings; failures retry from the outbox.

Aggregated operational data only. No medical records.

Patient — Mobile

Authentication, search, review and appointment management in a single-column flow.

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9:41

ClinicFlow

Patient

Welcome back

Email

alex@example.com

Password

.....

→] Sign in

New to ClinicFlow?

Create account

✓ Authorization continues at the API boundary.

9:41

ClinicFlow

Patient

Find care

Clinic

Downtown

Specialty

Dermatology

Date

Jul 14, 2026

🔍 Search availability

Available times

09:00

Available

10:30

Selected

11:00

Booked

9:41

ClinicFlow

Patient

Review booking

Selected slot

Dr. Maya Patel

Dermatology · Downtown Clinic
Tuesday, July 14 · 10:30–11:00

JUL 14

Tuesday · 10:30
30 minute appointment

Ready

⚠️ Reminder preference

Email and SMS requests queue after confirmation.

✓ Confirm appointment

9:41

ClinicFlow

Patient

My appointments

✓ Appointment confirmed

Jul 14 at 10:30 with Dr. Maya Patel.

JUL 14

Dermatology
10:30 · Downtown

Confirmed

Reschedule

Cancel appointment

✗ Slot unavailable

Refresh availability before selecting another time.

Reception — Mobile

Daily agenda, new appointment, calendar blocking and conflict recovery.

Draft for architect/product approval

9:41

ClinicFlow

Reception

Daily schedule

Doctor

Dr. Maya Patel

Tue, Jul 14 · 18 visits

08:30

Jordan Lee
Room 2

Confirmed

09:00

Sofia Rivera
Reminder pending

Pending

09:30

Team meeting
Doctor unavailable

Blocked

+ New appointment

9:41

ClinicFlow

Reception

New appointment

Patient

Sofia Rivera

Clinic

Downtown

Doctor

Dr. Maya Patel

10:30
Selected

11:00
Available

11:30
Booked

✓ Confirm booking

9:41

ClinicFlow

Reception

Block availability

Date

Jul 14, 2026

From

15:00

To

16:00

Reason

Clinic meeting

Calendar impact

Patients will see this period as blocked.

Add block

9:41

ClinicFlow

Reception

Resolve conflict

Slot no longer available

Another booking won the transaction.

Choose another available time

10:30
Booked

11:00
Selected

11:30
Available

Refresh availability

Save new time

Doctor — Mobile

Daily schedule, operational detail, empty and error states.

Draft for architect/product approval

9:41

ClinicFlow

Doctor

My schedule

Tue, July 14

08:30

Jordan Lee
New patient · Room 2

Checked in

09:00

Sofia Rivera
Follow-up · Room 2

Confirmed

09:30

Team meeting
Availability block

Blocked

10:00

Amir Hassan
Follow-up · Room 2

Confirmed

9:41

ClinicFlow

Doctor

Appointment detail

Privacy

No medical record data is stored in this view.

Checked in

Jordan Lee

08:30–09:00 · Room 2

Operational note

Patient requested accessible entrance directions.

✓ Mark completed

9:41

ClinicFlow

Doctor

My schedule

Calendar icon with X

No appointments today

Use date navigation to view another day.

Choose another date

9:41

ClinicFlow

Doctor

Schedule unavailable

Could not load schedule

Retry without exposing stale patient details.

Retry

Manager — Mobile

Operational analytics, clinic comparison, notification health and filters.

Draft for architect/product approval

9:41

ClinicFlow

Manager

Overview

Clinic

All clinics

Appointments

1,284

+8% this month

No-shows

76

5.9% of visits

Aggregated operational data only.

9:41

ClinicFlow

Manager

Clinic comparison

6.8%

Downtown Clinic

42 no-shows · 615 visits

Review

5.2%

Northside Clinic

21 no-shows · 402 visits

On track

4.9%

West Clinic

13 no-shows · 267 visits

On track

View no-shows

9:41

ClinicFlow

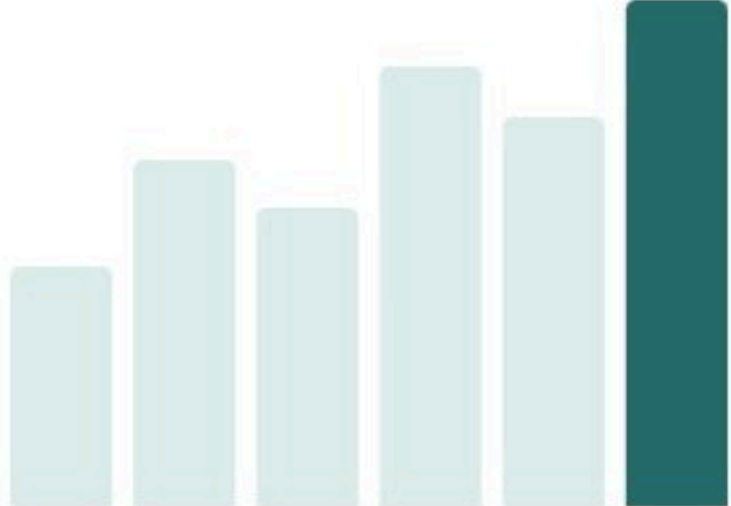
Manager

Notification health

98.7% delivered

14 pending retries

3 provider failures



Booking remains independent

Provider failures retry from the outbox.

9:41

ClinicFlow

Manager

Report filters

Clinic

All clinics

Date range

Jul 1–31, 2026

Status

All statuses

Specialty

All specialties

Apply filters

Clear filters